MagHeal : A Smart Magnetic Sleeve for Faster, Affordable Bone Healing

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Introduction

- 20% of fractures fail to heal after a prolonged period, causing non-union.
  - Symptoms can include persistent pain, swelling, and limited mobility
  - Physical therapy and use of brace offer temporary relief.
  - Osteotomy, an invasive surgery, can potentially heal non-union

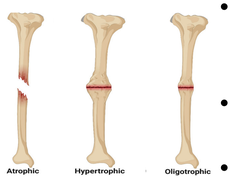


Fig 4: Three types of non-union fractures

$23,678
Estimated Average Cost for Osteotomy [6]

44%
Underinsured Americans [1]

26M
Uninsured Americans [1]

- Covering costs for surgery can cause a huge financial burden on many families.
- Healing time can take 3 months to a full year

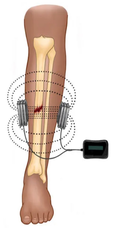


Fig 5: Pulsed EMF device healing tibial nonunion

- Research shows electromagnetic pulses mimic bioelectric signals that stimulate osteoblasts.
- In 15 trials with 1,247 patients, electrical stimulation reduced non-union risk by 35% compared to controls [2].
- Electromagnetic pulse therapy could improve non-union fracture treatment, making it more accessible.

Background

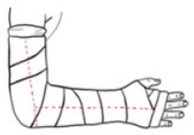


Fig 6: Casts wrap around the arm and hand for stability; this design will be used in the initial prototype.

- Pulses of hertz modulated into a trapezoidal waveform reduces rise time and elongates the pulse's duration [4].
- Patients wearing **1-30 Hz** pulses for at least **30 minutes** in the span of **6 weeks** showed signs of less pain and healing [2].

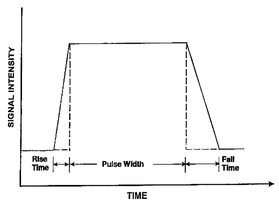
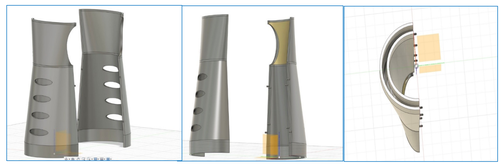
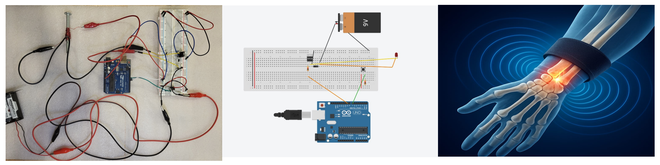


Fig 7: Example of a modulated magnetic pulse shaped into a trapezoidal waveform

Method



Mechanical Innovation
- Dual layer 3D-printed strap adjustable sleeve conforms to the forearm.
- keep fractures aligned like a cast and all electronics safely insulated from direct skin contact.
- Modelled using Fusion 360 software.
- lightweight, breathable, ergonomic brace.



Electrical Development
- Arduino UNO(Brain of our device) based PEMF generator.
- Coil wound around brace generates low-intensity fields.
- Hall Effect Sensor detects active magnetic fields.
- Circuit verified using TinkerCAD simulation.
- Microcontroller coded with Arduino IDE 2.3.6 to regulate the frequency.
- Digital display shows real-time frequency.

Solution

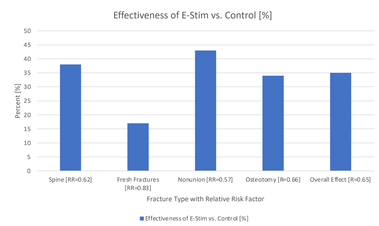


Fig 8: Each trial shows that patients prefer pulsed EMF therapy by the following percentages [5]

- Facilitates early-stage rehabilitation without surgical intervention.
- Designed for low-barrier adoption - no clinical infrastructure or technician required.
- Minimizes long-term care costs for uninsured or low-income users.

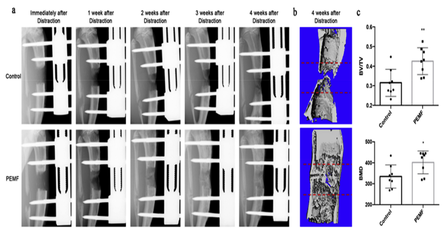


Fig 2: X-rays and 3D µCT show faster callus formation and higher bone density in PEMF group compared to control over 4 weeks.

- Combines medical-grade brace design with pulsed EMF therapy into a unified, wearable system.
- Offers a customizable healing protocol through frequency tuning, supporting individual patient needs.

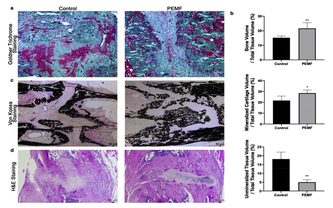


Fig. 1: Improved callus quality and bone consolidation in PEMF group, shown by Goldner Trichrome, H&E, and Von Kossa staining.

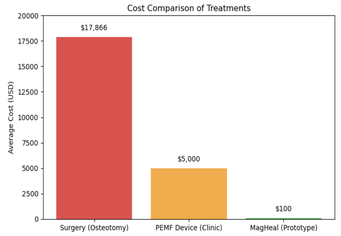


Fig. 3: Cost comparison of various treatments for Bone fracture

- Replaces passive immobilization (like casts) with an active healing approach using EMF stimulation.
- Enables on-demand therapy sessions with real-time feedback for confidence in use.
- Portable form factor allows treatment in any setting-at home, clinic, or on-the-go.
- Improves comfort and adherence by reducing the need for full immobilization.
- Provides visible proof-of-treatment (via display + feedback), which improves compliance.
- Flexible framework supports additional sensors for motion, temperature, or feedback logging.
- Aligned with FDA-supported PEMF healing frequencies from prior clinical research.
- Addresses global need for affordable fracture care with scalable, replicable design.

Impact

- Enables home recovery
- Non-invasive, affordable, portable
- Ideal for uninsured or underserved communities

Sustainability
- Minimal waste via 3D printing
- Reusable parts, low energy design
- Carbon-conscious Arduino sourcing



Future Work

- Integrate app for remote tracking + feedback.
- Improve power efficiency + safety features.
- Build our own microcontrollers using PCBs & embed multiple sensors.



Market Opportunity
- 3M+ annual fractures in the U.S.
- Market in sports rehab, rural clinics, and telehealth

Acknowledgement

- The project was funded by Innovation Studio and our group.
- Special thanks to Brad Eshelman for helping us 3D print our arm brace and ECE help desk for support with circuits.

References

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[6] Mdsave.com, 2024.